

Dee Aein

Newark, CA | +1(510)-980-3271 | deeaen.ts@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

San Jose State University

Aug 2025 - Dec 2027

Bachelor of Science, Computer Science (GPA: 3.93)

San Jose, CA

- **Coursework:** Data Structures & Algorithms, Database Systems, Linear Algebra, Graph Theory, Object Oriented Programming, Advanced Python, Artificial Intelligence, Machine Learning (regression, classification, MLP, CNN, autoencoders, attention mechanism, model evaluation)

SKILLS

- **Languages:** Java, Python, SQL, JavaScript, C
- **Technologies:** REST APIs, FastAPI, Flask, PostgreSQL, MySQL, SQLAlchemy, Alembic, Docker, AWS (ECS/Fargate, RDS, S3, CloudFront, ECR, ALB, IAM), RAG, Sentence Transformers, OpenAI API, scikit-learn, NumPy, pandas, Matplotlib, JavaFX, Java Servlets, Git, GitHub, Maven

PROJECT EXPERIENCE

AI Codebase Assistant (RAG)

Jul 2026 – Present

Python, JavaScript, FastAPI, PostgreSQL, Docker, AWS ECS/Fargate, RDS, S3, HTML, CSS, SQLAlchemy (ORM), pgvector, OpenAI API, Sentence Transformers, Pydantic

- Engineered an **end-to-end repository ingestion pipeline** that processed 55 source files into 158 searchable code chunks in 27 seconds by integrating the GitHub API, file filtering, AST-based chunking, embeddings, and PostgreSQL storage.
- Achieved **100% top-10 and 86.7% top-2 retrieval success** across 15 repository-specific questions on 2 codebases by implementing semantic retrieval with Sentence Transformer embeddings and pgvector cosine-similarity search.
- Achieved **100% grounded responses** across 15 manually evaluated answers by constraining OpenAI-generated responses to retrieved code context and returning exact file- and line-level citations through REST API endpoints.
- **Deployed a reproducible full-stack application to AWS** by containerizing services with **Docker** and integrating ECR, ECS/Fargate, RDS PostgreSQL, S3, CloudFront, an Application Load Balancer, IAM, Secrets Manager, and security-group-based networking.

MedIMS – 3-Tier Inventory Management System

Mar 2026 - Apr 2026

Java, Java Servlets, MySQL, JDBC, BCrypt, AWS

- **Led a 3-person team** to deliver a role-based hospital inventory system on schedule by coordinating development workflows, delegating tasks, and driving project execution
- Designed a **normalized MySQL data layer across 15 relational tables** by translating ER diagrams into DAO-based persistence logic with complex JOIN queries and prepared statements
- Built backend services across **11 REST endpoints** by implementing Java Servlet for CRUD operations, submission/approval workflows, session-based authentication, and BCrypt password hashing.

Snake Game – Real-Time Interactive Application

Oct 2025 - Nov 2025

Java, JavaFx, OOP, Event-Driven Programming

- Engineered a **real-time JavaFx game sustaining 50 FPS** by implementing an AnimationTimer game loop, collision detection, and 20-pixel grid-based state management
- Improved movement reliability and state consistency by modeling snake segments with **custom SnakeBody objects** and **dynamic list updates**, preventing invalid reverse-direction inputs and maintaining synchronized navigation and scoring state.

Rental Marketplace System – CLI-Based Backend Simulation

Aug 2025 - Sep 2025

Java, OOP, HashMap, Exception Handling

- **Led a 5-person team** to build a role-based rental platform supporting 9 vendor and customer operations by coordinating development and implementing inventory, pricing, rental, and return workflows over 150+ in-memory product instances.
- Prevented identifier collisions and enforced consistent business rules across by designing a modular object-oriented architecture with an abstract Product class, Rentable/Buyable interfaces, composite-key mapping and exception-based validations.

WORK EXPERIENCE

Math, Science & Technology Tutor

Sep 2023 - Jan 2025

De Anza college

- Led weekly tutoring support for 30+ students across math and technical subjects, **applying data structures, algorithms, and structured reasoning** to strengthen problem-solving methods to strengthen analytical skills
- Mentored 2–3 students per quarter over 1.5 years through recurring 1:1 sessions by evaluating progress, adapting instruction, and guiding students toward stronger performance in STEM coursework

ICC Representative, Competitive Programming Club

· Mar 2024 - Aug 2025

De Anza College

- Represented the Competitive Programming club in Inter-Club Council meetings over 1.5 years, **coordinating with campus organizations and administrators** to communicate club initiatives and operational requirements
- Led weekly data structures and algorithm problem-solving sessions for ~10 students, guiding participants from brute-force solutions to optimized data-structure and algorithm approaches while strengthening time-complexity analysis.